

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12388346
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Yang; Shang-Kay et al.

Power converter with reduced power consumption and method of operating the same

Abstract

A power converter with reduced power consumption receives an input voltage and provides an output voltage to supply power to a load. The power converter includes a power factor correction circuit, a driver circuit, and a controller. The power factor correction circuit includes at least one inductor and an output capacitor. When the controller determines that an effective value of an input current is correspondingly lower than a current lower limit, a specific time period starts based on the output voltage reaching a voltage lower limit, and the specific time period ends based on the output voltage reaching a voltage upper limit. In the specific time period, the controller enables the driver circuit based on the input current being correspondingly greater than a predetermined threshold so as to continuously transfer the energy stored in the at least one inductor to the output capacitor.

Inventors: Yang; Shang-Kay (Taoyuan, TW), Wang; Hsien-Kai (Taoyuan, TW), Lin; Yen-Wei (Taoyuan, TW)

Applicant: DELTA ELECTRONICS, INC. (Taoyuan, TW)

Family ID: 91162265

Assignee: DELTA ELECTRONICS, INC. (Taoyuan, TW)

Appl. No.: 18/300190

Filed: April 13, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240178740 A1	May. 30, 2024

Foreign Application Priority Data

CN	202211490928.3	Nov. 25, 2022
----	----------------	---------------

Publication Classification

Int. Cl.: H02M1/08 (20060101); H02M1/00 (20060101); H02M1/42 (20070101)

U.S. Cl.:

CPC H02M1/0054 (20210501); H02M1/0035 (20210501); H02M1/08 (20130101); H02M1/4233 (20130101);

Field of Classification Search

CPC: H02M (1/0054); H02M (1/0035); H02M (1/08); H02M (1/4233)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2017/0302157	12/2016	Lin et al.	N/A	N/A
2020/0335983	12/2019	Long et al.	N/A	N/A
2022/0131470	12/2021	Sagneri et al.	N/A	N/A
2024/0195291	12/2023	Yang	N/A	H02M 1/4208

OTHER PUBLICATIONS

Office Action dated Nov. 21, 2023 of the corresponding Taiwan patent application No. 111145126. cited by applicant

Primary Examiner: Berhane; Adolf D

Attorney, Agent or Firm: MUNCY, GEISSLER, OLDS & LOWE, P.C.

Background/Summary

BACKGROUND

Technical Field

(1) The present disclosure relates to a power converter and a method of operating the same, and more particularly to a power converter with reduced power consumption and a method of operating the same.

Description of Related Art

(2) The statements in this section merely provide background information related to the present disclosure and do not necessarily constitute prior art.

(3) Due to the rapid development of the information industry in recent years, the power supply plays an important role, and the power consumption of information or home appliances is gradually required to be reduced to a certain value under no-load operation or light-load operation. On the other hand, since harmonic suppression has classified limits in the international standard IEC61000-3-2, and stipulates the maximum rated harmonic current, information or home appliances must be equipped with power converters with power factor correction functions so as to

improve the power factor of the AC power supply.

(4) Since the general power converter with power factor correction function is under no load, in order to maintain the output voltage, the magnitude of the PWM signal follows the control value of the controller (as shown in FIG. 1). When the instantaneous value of the input voltage appears at a lower input voltage V_{in} (usually near the zero-crossing point), the energy transferred will be very low (i.e., $V_{in}(wt) \cdot I_{in}$). The power converter needs to output more pulses (corresponding to the inductor current I_L) to stabilize the output voltage V_{out} , resulting in increased driving loss of the power converter. Similarly, since the power converter needs to output more pulses when the instantaneous value of the input voltage appears at a lower input voltage V_{in} , the conversion efficiency of the current power converter will be poor, and additional power consumption will be increased.

(5) Accordingly, a power converter with reduced power consumption and a method of operating the same are provided to improve the conversion efficiency of the power factor correction circuit and reduce the driving loss of the power converter under no-load operation or light-load operation.

SUMMARY

(6) In order to solve the above-mentioned problems, the preset disclosure provides a power converter with reduced power consumption. The power converter with reduced power consumption includes a power factor correction circuit, a driver circuit, and a controller. The power factor correction circuit includes at least one inductor, at least one power switch, and an output capacitor, and receives the input voltage. The driver circuit is coupled to the at least one power switch. The controller is coupled to the driver circuit, and provides a PWM signal to the driver circuit. The driver circuit drives the switching of the at least one power switch based on the PWM signal so as to control the power factor correction circuit to convert the input voltage into the output voltage. When the controller determines that an effective value of an input current of the power factor correction circuit is correspondingly lower than a current lower limit, a specific time period starts based on the output voltage reaching a voltage lower limit, and the specific time period ends based on the output voltage reaching a voltage upper limit. In the specific time period, the controller enables the driver circuit based on the input current being correspondingly greater than a predetermined threshold so as to operate the switching of the at least one power switch through the PWM signal.

(7) In order to solve the above-mentioned problems, the preset disclosure provides a method of operating a power converter with reduced power consumption. The power converter receives an input voltage and provides an output voltage to supply power to a load. The power converter includes a power factor correction circuit and a driver circuit, and the power factor correction circuit includes at least one inductor, at least one power switch, and an output capacitor. The method includes steps of: (a) determining that an effective value of an input current of the power factor correction circuit is correspondingly less than a current lower limit; (a1) starting a specific time period when the output voltage reaches a voltage lower limit; (a11) enabling the driver circuit when the input current is correspondingly greater than a predetermined threshold to provide a PWM signal to drive the switching of the at least one power switch; (a12) disabling the driver circuit when the input current is correspondingly lower than the predetermined threshold; and (a2) ending the specific time period when the output voltage reaches a voltage upper limit.

(8) The main purpose and function of the present disclosure is to provide a method for controlling a conduction phase angle interval of the input current. When the load is light to a certain extent (i.e., no load or light load), the method of burst control of the output voltage (i.e., the burst mode) is used to make the input current reach near the peak value of the input voltage so that when the input current reaches near the peak value of the input voltage, the driver circuit is enabled. Therefore, the controller may only operate the switching of the power switches when the driver circuit is enabled so that the energy stored in the inductor can be effectively transferred to the output capacitor. Especially when the power factor correction circuit is under no-load condition or light-load

condition, and the input current is not high, the effect is even more remarkable. Therefore, the power converter and its operating method of the present disclosure can achieve the effect of increasing the efficiency of the power factor correction circuit and reducing the driving loss of the power factor correction circuit when it is under no-load operation or light-load operation.

(9) It is to be understood that both the foregoing general description and the following detailed description are exemplary, and are intended to provide further explanation of the present disclosure as claimed. Other advantages and features of the present disclosure will be apparent from the following description, drawings, and claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawing as follows:
- (2) FIG. 1 is a waveform diagram of power factor correction of a conventional power converter.
- (3) FIG. 2A is a block circuit diagram of a power converter with reduced power consumption according to the present disclosure.
- (4) FIG. 2B is a schematic diagram of power consumption reduction of the power converter according to the present disclosure.
- (5) FIG. 3 is a block circuit diagram of a controller according to the present disclosure.
- (6) FIG. 4 is an operation sequence diagram of the power converter according to the present disclosure.
- (7) FIG. 5 is a curve diagram of a conduction phase angle interval of an input current according to the present disclosure.
- (8) FIG. 6A is a flowchart of a method of operating the power converter with reduced power consumption in an interval I according to the present disclosure.
- (9) FIG. 6B is a flowchart of the method of operating the power converter with reduced power consumption during an interval II and an interval III according to the present disclosure.

DETAILED DESCRIPTION

- (10) Reference will now be made to the drawing figures to describe the present disclosure in detail. It will be understood that the drawing figures and exemplified embodiments of present disclosure are not limited to the details thereof.
- (11) Please refer to FIG. 2A, which shows a block circuit diagram of a power converter with reduced power consumption according to the present disclosure, and also refer to FIG. 1. The power converter **100** receives an input voltage V_{in} and provides an output voltage V_{out} to supply power to a load **200**. The power converter **100** includes a power factor correction circuit **1**, a driver circuit **2**, and a controller **3**. The power factor correction circuit **1** includes at least one inductor L , at least one switch (SH , SL , $S1$, $S2$), and an output capacitor C_{out} . Take a circuit structure of FIG. 2A as an example, the inductor L is coupled at an input end of the power factor correction circuit **1** and receives the input voltage V_{in} . The output capacitor C_{out} is coupled at an output end of the power factor correction circuit **1** and coupled to the load **200** to provide the output voltage V_{out} to supply power to the load **200**. The power switches (SH , SL , $S1$, $S2$) are coupled between the inductor L and the output capacitor C_o , and the driver circuit **2** is coupled between the controller **3** and the power switches (SH , SL , $S1$, $S2$).
- (12) The power converter **100** further includes a voltage detection circuit **4** and a current detection circuit **5**. The voltage detection circuit **4** is coupled between the output capacitor C_{out} and the controller **3**, and the current detection circuit **5** is coupled between the power factor correction circuit **1** and the controller **3**. The voltage detection circuit **4** detects the output voltage V_{out} to provide a first feedback voltage V_{fb} corresponding to the output voltage V_{out} to the controller **3**.

The current detection circuit 5 detects an input current I_{in} flowing through the input end of the power factor correction circuit 1 to provide a second feedback voltage I_{fb} corresponding to the input current I_{in} to the controller 3. The controller 3 modulates a PWM (pulse-width modulation) signal PWM based on the first feedback voltage V_{fb} and the second feedback voltage I_{fb} and provides the PWM signal PWM to the driver circuit 2.

(13) The driver circuit 2 receives the PWM signal PWM and drives power switches (SH, SL, S1, S2) to switch turning on and turning off based on the PWM signal PWM. Therefore, the driver circuit 2 drives the switching of the power switches (SH, SL, S1, S2) based on the PWM signal PWM so as to control the power factor correction circuit 1 converting the input voltage V_{in} into the output voltage V_{out} . In one embodiment, the circuit structure of the power factor correction circuit 1 shown in FIG. 2A is only a schematic example. The power factor correction circuit 1 may use different circuit structures according to the requirements of the power converter 100. Therefore, all AC/DC conversion circuits that may be used as the power factor correction circuit 1 should be included in the scope of the present embodiment, and the detail description is omitted here for conciseness.

(14) Please refer to FIG. 2B, which shows a schematic diagram of power consumption reduction of the power converter according to the present disclosure, and also refer to FIG. 2A. With respect to the related art of FIG. 1, if the power factor correction circuit 1 can reach or near the highest (peak) point of the mains voltage (that is, the instantaneous voltage value of the input voltage V_{in} is the peak value), and then transfer the energy to the output end, not only the efficiency of the power factor correction circuit 1 can be increased, but also the driving loss of the power factor correction circuit 1 at no load or light load can be reduced. Therefore, the main purpose and function of the present disclosure is to provide a method for controlling the conduction phase angle interval ϕ of the input current I_{in} . The conduction phase angle interval ϕ takes the peak value of the input voltage V_{in} as the center to set the predetermined threshold I_{in_min} . When the load is light to a certain extent (i.e., no load or light load), the method of burst control of the output voltage V_{out} (i.e., the burst mode) is used to make the input current I_{in} reach near the peak value of the input voltage V_{in} so that when the input current I_{in} reaches near the peak value of the input voltage V_{in} , the driver circuit 2 is enabled, and the driver circuit 2 is disabled for the rest of the time.

(15) Therefore, the controller 3 may only operate the switching of the power switches (SH, SL, S1, S2) when the driver circuit 2 is enabled so that the energy stored in the inductor L can be effectively transferred to the output capacitor C_{out} . In other words, when the input current I_{in} is greater than the predetermined threshold I_{in_min} , the inductor L generates the inductor current I_L to transfer the energy stored in the inductor L to the output capacitor C_{out} , and during the period when the remaining input current I_{in} is lower than the predetermined threshold I_{in_min} , there is no generation of inductor current. Therefore, the energy transfer of the inductor L may be made more efficient, especially when the power factor correction circuit 1 is under no-load condition or light-load condition, and the input current I_{in} is not high, the effect is even more remarkable. Therefore, the power converter 100 and its operating method of the present disclosure can achieve the effect of increasing the efficiency of the power factor correction circuit 1 and reducing the driving loss of the power factor correction circuit 1 when it is under no-load operation or light-load operation.

(16) Please refer to FIG. 3, which shows a block circuit diagram of a controller according to the present disclosure, and also refer to FIG. 2A and FIG. 2B. The controller 3 includes an error amplifier 32, a voltage controller 34, and a current controller 36. The error amplifier 32 is coupled to the output capacitor C_{out} through the voltage detection circuit 4 to receive the first feedback voltage V_{fb} corresponding to the output voltage V_{out} . The error amplifier 32 generates an error signal V_{er} based on the first feedback voltage V_{fb} and a reference voltage V_{ref} . The voltage controller 34 is coupled to the error amplifier 32 to receive the error signal V_{er} . The voltage controller 34 generates a voltage command V_c based on the error signal V_{er} . The current controller 36 is coupled to the voltage controller 34 to receive the voltage command V_c . The current

controller **36** modulates and generates the PWM signal PWM to provide the PWM signal PWM to the driver circuit **2**. Therefore, the frequency/duty ratio of the PWM signal PWM is mainly determined by the voltage command V_c , and the driver circuit **2** operates the switching of the power switches (SH, SL, S1, S2) based on the PWM signal PWM.

(17) Furthermore, corresponding to the purpose and effect mentioned in the above, the current controller **36** is further coupled to the current detection circuit **5** to receive the second feedback voltage I_{fb} corresponding to the input current I_{in} . The current controller **36** generates a first current command $I_{in_rms_ref}$ corresponding to an effective value of the input current I_{in} based on the second feedback voltage I_{fb} (corresponding to the input current I_{in}). The first current command $I_{in_rms_ref}$ is proportional to the effective value of the input current I_{in} (that is, a root-mean-square value of the input current I_{in}), and is combined with the voltage command V_c as a DC current control command for the next cycle. The current controller **36** further generates a second current command I_{in_ref} corresponding to a current waveform of the input current I_{in} based on the second feedback voltage I_{fb} (corresponding to the input current I_{in}). The second current command I_{in_ref} is proportional to an instantaneous value of the input current I_{in} , and is combined with the voltage command V_c as an AC sinusoidal current control command for the next cycle. Based on conservation of energy, by sampling the output voltage V_{out} and the input current I_{in} , the loading of the back-end load **200** can be reflected, and therefore it can be determined whether the loading of the load **200** is at no load (light load), medium load, or heavy load. In addition, the controller **3** can preset parameters such as the current lower limit, the current upper limit, and the predetermined threshold based on the circuit specifications of the power factor correction circuit **1** for subsequent operation of the driver circuit **2**.

(18) Therefore, the current controller **36** can determine whether to enable or disable the driver circuit **2** by providing the control signal S_c through the voltage command V_c , the second feedback voltage I_{fb} , and the parameters set above. In one embodiment, the current lower limit, the current upper limit, and the predetermined threshold may preferably be preset by the current controller **36** based on the voltage command V_c , the second feedback voltage I_{fb} , and the circuit specifications of the power factor correction circuit **1**, but not limited to this.

(19) Please refer to FIG. 4, which shows an operation sequence diagram of the power converter according to the present disclosure, and also refer to FIG. 2A to FIG. 3. In FIG. 4, three intervals (I, II, III) are shown, and the three intervals respectively correspond to the timings when the power converter **100** operates at no load (light load), medium load, and heavy load. The controller **3** presets parameters such as the current lower limit $I_{in_rms_min}$, the current upper limit $I_{in_rms_max}$, and the predetermined threshold I_{in_min} based on the circuit specifications of the power factor correction circuit **1**. The current controller **36** mainly determines whether the effective value of the input current I_{in} is correspondingly lower than the current lower limit $I_{in_rms_min}$ through the first current command $I_{in_rms_ref}$, and determines whether the instantaneous value of the input current I_{in} is correspondingly greater than the predetermined threshold I_{in_min} so as to perform corresponding operations.

(20) In the interval I, the current controller **36** determines that the effective value of the input current I_{in} of the power factor correction circuit **1** is correspondingly lower than the current lower limit $I_{in_rms_min}$ based on the first current command $I_{in_rms_ref}$, that is, it indicates that the power factor correction circuit **1** operates under no-load condition or light-load condition. Therefore, the controller **3** operates in a burst mode. The operation of the burst mode is that the controller **3** controls the PWM signal PWM to generate pulses in a specific time period T_1 , and does not generate pulses in other time periods so that the power converter **100** save power consumption when operating in the no-load or the light-load condition. In one embodiment, the manner of setting the start/end of the specific time period T_1 is based on the magnitude of the output voltage V_{out} . When the output voltage V_{out} reaches a voltage lower limit V_{out_min} , the specific time period T_1 starts, and when the output voltage V_{out} reaches a voltage upper limit

Vout_max, the specific time period T1 ends.

(21) Also refer to FIG. 2B, in the specific time period T1, when the current controller 36 of the controller 3 determines that the instantaneous value of the input current I_{in} of the power factor correction circuit 1 is correspondingly greater than the predetermined threshold I_{in_min} through the second current command I_{in_ref} , the driver circuit 2 is enabled. When the driver circuit 2 is enabled, the driver circuit 2 drives the power switches (SH, SL, S1, S2) based on the pulse wave of the PWM signal PWM to operate the switching of the power switches (SH, SL, S1, S2) so that the energy stored in the inductor L is transferred to the output capacitor Cout. Conversely, when the current controller 36 determines that the instantaneous value of the input current I_{in} of the power factor correction circuit 1 is correspondingly lower than the predetermined threshold I_{in_min} through the second current command I_{in_ref} , the driver circuit 2 is disabled. Therefore, the pulse wave of the PWM signal PWM cannot drive the power switches (SH, SL, S1, S2) since the driver circuit 2 is disabled.

(22) Furthermore, the current controller 36 of the controller 3 actually continuously generates the PWM signal PWM based on the voltage command Vc. Especially in the burst mode in the interval I, the PWM signal PWM is continuously generated only in the specific time period T1. That is, in the general burst mode, the pulse wave of the PWM signal PWM will actually be more than the pulse wave in the specific time period T1 in FIG. 4. In the general burst mode, besides the pulse wave located near the peak of the waveform of the input voltage V_{in} , the pulse wave located near the zero point of the waveform of the input voltage V_{in} is also included. However, as shown in FIG. 2B, when the load is light enough to a certain extent (i.e., no load or light load), the burst mode of the output voltage Vout is combined to make the input current I_{in} reach the peak value of the waveform of the input voltage V_{in} , and be greater than the predetermined threshold I_{in_min} , the driver circuit 2 is enabled. Therefore, in FIG. 4, only during the specific time period T1 and the input current I_{in} (current waveform) is correspondingly greater than the predetermined threshold I_{in_min} , the switching of the power switches (SH, SL, S1, S2) is operated to make the energy stored in the inductor L can be effectively transferred to the output capacitor Cout to correspondingly generate the input current I_{in} .

(23) In the interval II, the current controller 36 determines that the effective value of the input current I_{in} of the power factor correction circuit 1 is between the current lower limit $I_{in_rms_min}$ and the current upper limit $I_{in_rms_max}$ through the first current command $I_{in_rms_ref}$, that is, it indicates that the power factor correction circuit 1 operates from no-load condition or light-load condition to medium-load condition. In this condition, the controller 3 leaves from the operation of the burst mode, the ripple of the output voltage Vout becomes smaller, and will no longer reach the voltage lower limit Vout_min and voltage upper limit Vout_max.

(24) Since the controller 3 leaves from the operation of the burst mode, the enabling/disabling of the driver circuit 2 is no longer limited to the specific time period T1. When the current controller 36 of the controller 3 determines that the instantaneous value of the input current I_{in} of the power factor correction circuit 1 is correspondingly lower than the predetermined threshold I_{in_min} through the second current command I_{in_ref} , the driver circuit 2 is disabled, and the pulse wave of the PWM signal PWM cannot drive the power switches (SH, SL, S1, S2) since the driver circuit 2 is disabled. Conversely, when the current controller 36 determines that the instantaneous value of the input current I_{in} of the power factor correction circuit 1 is correspondingly greater than the predetermined threshold value I_{in_min} through the second current command I_{in_ref} , the driver circuit 2 is enabled. When the driver circuit 2 is enabled, the driver circuit 2 drives the power switches (SH, SL, S1, S2) based on the pulse wave of the PWM signal PWM to operate the switching of the power switches (SH, SL, S1, S2) so that the energy stored in the inductor L is transferred to the output capacitor Cout.

(25) In the interval III, the current controller 36 determines that the effective value of the input current I_{in} of the power factor correction circuit 1 is correspondingly greater than the current upper

limit $I_{in_rms_max}$ through the first current command $I_{in_rms_ref}$, that is, it indicates that the power factor correction circuit **1** operates from medium-load condition to heavy-load condition. Since the power factor correction circuit **1** operates under heavy load, the energy required by the load **200** is relatively large so that the current controller **36** of the controller **3** continuously enables the driver circuit **2**. The driver circuit **2** continuously drives the power switches (SH, SL, S1, S2) based on the pulse wave of the PWM signal PWM to operate the switching of the power switches (SH, SL, S1, S2) so that the energy stored in the inductor L is continuously transferred to the output capacitor Cout.

(26) Please refer to FIG. 5, which shows a curve diagram of a conduction phase angle interval of an input current according to the present disclosure, and also refer to FIG. 2A to FIG. 4. The intervals I, II, and III in FIG. 5 correspond to the intervals I, II, and III in FIG. 4 respectively, and the curve in FIG. 5 refers to the width of the conduction phase angle interval ϕ in FIG. 2B. In the interval I, since the effective value of the input current I_{in} of the power factor correction circuit **1** is correspondingly lower than the current lower limit $I_{in_rms_min}$, the conduction phase angle interval ϕ is maintained at a minimum interval ϕ_{min} . In the interval II, since the load gradually increases, the effective value of the input current I_{in} of the power factor correction circuit **1** is between the current lower limit $I_{in_rms_min}$ and the current upper limit $I_{in_rms_max}$ so that the conduction phase angle interval ϕ is between the minimum interval ϕ_{min} and a maximum interval ϕ_{max} . In the interval III, due to the heavy load, the effective value of the input current I_{in} of the power factor correction circuit **1** is correspondingly greater than the current upper limit $I_{in_rms_max}$ so that the conduction phase angle interval ϕ is maintained at the maximum interval ϕ_{max} . In particular, the maximum interval ϕ_{max} usually refers to the interval in which the driver circuit **2** is continuously enabled, which means that the interval may be 180 degrees; the minimum interval ϕ_{min} may be designed according to the specifications of the power factor correction circuit **1** (such as but not limited to 15 degrees).

(27) Please refer to FIG. 6A, which is a flowchart of a method of operating the power converter with reduced power consumption in an interval I according to the present disclosure; please refer to FIG. 6B, which is a flowchart of the method of operating the power converter with reduced power consumption during an interval II and an interval III according to the present disclosure, and also refer to FIG. 2A to FIG. 5. The present disclosure mainly uses a method of controlling the conduction phase angle interval ϕ of the input current I_{in} to control the power factor correction circuit **1**. The driver circuit **2** is enabled only when the input current I_{in} reaches near the peak value of the waveform of the input voltage V_{in} , and the driver circuit **2** is disabled at other times. Therefore, the method of operating the power converter includes steps of generating a first current command corresponding to an effective value of the input current and a second current command corresponding to a current waveform of the input current (S100). In particular, the method of generating the first current command $I_{in_rms_ref}$ and the second current command I_{in_ref} may be referred to FIG. 3, and the detail description is omitted here for conciseness.

(28) Afterward, determining whether the effective value of the input current is correspondingly greater than a current upper limit (S120). When the effective value of the input current I_{in} is correspondingly greater than the current upper limit $I_{in_rms_max}$, the operation in the interval III in FIG. 4 is executed, and the operation of the burst mode leaves (S200) and the driver circuit is continuously enabled (S220). Detailed operations of the above steps may be referred to the interval III in FIG. 4 and FIG. 5, and the detail description is omitted here for conciseness. When the determination result in step (S120) is "NO", step (S140) is then executed to determine whether the effective value of the input current is correspondingly lower than a current lower limit. When the effective value of the input current I_{in} is neither greater than the current upper limit $I_{in_rms_max}$ nor lower than the current lower limit $I_{in_rms_min}$, that is, when the determination result in step (S140) is "NO", the operation in the interval II in FIG. 4 is executed, and the operation of the burst mode leaves (S300).

(29) Afterward, determining whether the input current is correspondingly lower than a predetermined threshold (S320). When the instantaneous value of the input current I_{in} (current waveform) of the power factor correction circuit 1 is correspondingly lower than the predetermined threshold I_{in_min} , the driver circuit is disabled (S340). The pulse wave of the PWM signal PWM cannot drive the power switches (SH, SL, S1, S2) since the driver circuit 2 is disabled. Conversely, when the instantaneous value of the input current I_{in} (current waveform) of the power factor correction circuit 1 is correspondingly greater than the predetermined threshold I_{in_min} , the driver circuit is enabled (S360). When the driver circuit 2 is enabled, the driver circuit 2 can drive the power switches (SH, SL, S1, S2) based on the pulse wave of the PWM signal PWM to operate the switching of the power switches (SH·SL·S1·S2) through the PWM signal PWM so that the energy stored in the inductor L is continuously transferred to the output capacitor Cout.

(30) Afterward, when the determination result in step (S140) is “YES”, step (S400) is then executed to set the first current command as the current lower limit and to operate in the burst mode. The reason for setting the first current command $I_{in_rms_ref}$ to the current lower limit $I_{in_rms_min}$ is to avoid the first current command $I_{in_rms_ref}$ from being too low and causing the current controller 36 to malfunction. Afterward, determining whether the output voltage reaches a voltage upper limit (S420). When the output voltage Vout does not reach the voltage upper limit Vout_max, it is determined whether the output voltage reaches a voltage lower limit (S440). When the output voltage Vout reaches the voltage upper limit Vout_max, the specific time period ends (S460). When the output voltage Vout reaches the voltage lower limit Vout_min, the specific time period starts (S480).

(31) Afterward, determining whether in the specific time period (S500). When it is determined that it is not in the specific time period T1, the driver circuit is disabled (S520), and the pulse wave of the PWM signal PWM cannot drive the power switches (SH, SL, S1, S2) since the driver circuit 2 is disabled. When it is determined that it is in the specific time period T1, it is determined whether the input current is correspondingly lower than the predetermined threshold (S540). When it is determined that the instantaneous value of the input current I_{in} (current waveform) of the power factor correction circuit 1 is correspondingly lower than the predetermined threshold I_{in_min} , step (S520) is then executed, and the pulse wave of the PWM signal PWM cannot drive the power switches (SH·SL·S1·S2) since the driver circuit 2 is disabled. Conversely, when the determination result in step (S540) is “NO”, the driver circuit is enabled (S560). When the driver circuit 2 is enabled, the driver circuit 2 can drive the power switches (SH, SL, S1, S2) based on the pulse wave of the PWM signal PWM so as to operate the switching of the power switches (SH·SL·S1·S2) through the PWM signal PWM so that the energy stored in the inductor L is continuously transferred to the output capacitor Cout. In one embodiment, the process steps not described in FIG. 6A and FIG. 6B may be referred to in conjunction with FIG. 3 to FIG. 5, and the detail description is omitted here for conciseness.

(32) Although the present disclosure has been described with reference to the preferred embodiment thereof, it will be understood that the present disclosure is not limited to the details thereof. Various substitutions and modifications have been suggested in the foregoing description, and others will occur to those of ordinary skill in the art. Therefore, all such substitutions and modifications are intended to be embraced within the scope of the present disclosure as defined in the appended claims.

Claims

1. A power converter with reduced power consumption, configured to receive an input voltage and provide an output voltage to supply power to a load, the power converter comprising: a power factor correction circuit, comprising at least one inductor, at least one power switch, and an output

capacitor, and configured to receive the input voltage, a driver circuit, coupled to the at least one power switch, and a controller, coupled to the driver circuit, and configured to provide a PWM signal to the driver circuit; wherein the driver circuit drives the switching of the at least one power switch based on the PWM signal so as to control the power factor correction circuit to convert the input voltage into the output voltage, wherein when the controller determines that an effective value of an input current of the power factor correction circuit is correspondingly lower than a current lower limit, a specific time period starts based on the output voltage reaching a voltage lower limit, and the specific time period ends based on the output voltage reaching a voltage upper limit; in the specific time period, the controller enables the driver circuit based on the input current being correspondingly greater than a predetermined threshold so as to operate the switching of the at least one power switch through the PWM signal.

2. The power converter as claimed in claim 1, wherein during the specific time period, the controller disables the driver circuit based on the input current being correspondingly lower than the predetermined threshold.
3. The power converter as claimed in claim 1, wherein the controller continuously generates the PWM signal, and shields the PWM signal continuously generated by the controller by disabling the driver circuit.
4. The power converter as claimed in claim 1, wherein the controller continuously enables the driver circuit when determining that the effective value is correspondingly greater than a current upper limit so as to operate the at least one power switch by the PWM signal.
5. The power converter as claimed in claim 4, wherein the controller disables the driver circuit when determining the effective value is between the current lower limit and the current upper limit based on the input current being correspondingly lower than the predetermined threshold; the controller enables the driver circuit based on the input current being correspondingly greater than the predetermined threshold to provide the PWM signal to operate the at least one power switch.
6. The power converter as claimed in claim 4, wherein the controller comprises: an error amplifier, coupled to the output capacitor, and configured to generate an error signal based on the output voltage and a reference voltage, a voltage controller, coupled to the error amplifier, and configured to generate a voltage command corresponding to the PWM signal based on the error signal, and a current controller, configured to generate the PWM signal based on the voltage command, wherein the controller is configured to set the current lower limit, the current upper limit, and the predetermined threshold; the current controller is configured to generate a first current command corresponding to the effective value based on the voltage command and the input current, and generate a second current command corresponding to the input current based on the voltage command and the input current.
7. The power converter as claimed in claim 6, wherein the controller is configured to determine whether the effective value is correspondingly lower than the current lower limit based on the first current command, and determine whether the input current is correspondingly greater than the predetermined threshold based on the second current command.
8. The power converter as claimed in claim 6, further comprising: a voltage detection circuit, coupled to the output capacitor, and configured to detect the output voltage so as to provide a first feedback voltage, and a current detection circuit, coupled to an input end of the power factor correction circuit, and configured to detect the input current to provide a second feedback voltage, wherein the error amplifier is configured to generate the error signal based on the first feedback voltage and the reference voltage, and the current controller is configured to generate the first current command and the second current command based on the second feedback voltage and the voltage command.
9. A method of operating a power converter with reduced power consumption, the power converter configured to receive an input voltage and provide an output voltage to supply power to a load; the power converter comprising a power factor correction circuit and a driver circuit, and the power

factor correction circuit comprising at least one inductor, at least one power switch, and an output capacitor; the method comprising steps of: (a) determining that an effective value of an input current of the power factor correction circuit is correspondingly less than a current lower limit, (a1) starting a specific time period when the output voltage reaches a voltage lower limit, (a11) enabling the driver circuit when the input current is correspondingly greater than a predetermined threshold to provide a PWM signal to drive the switching of the at least one power switch, (a12) disabling the driver circuit when the input current is correspondingly lower than the predetermined threshold, and (a2) ending the specific time period when the output voltage reaches a voltage upper limit.

10. The method of operating the power converter as claimed in claim 9, further comprising steps of: (b) determining that the effective value is correspondingly greater than a current upper limit, and (b1) continuously enabling the driver circuit to operate the least one power switch by the PWM signal.

11. The method of operating the power converter as claimed in claim 9, further comprising steps of: (c) determining that the effective value is between the current lower limit and the current upper limit, (c1) enabling the driver circuit when the input current is correspondingly greater than the predetermined threshold, and (c12) disabling the driver circuit when the input current is correspondingly lower than the predetermined threshold.

12. The method of operating the power converter as claimed in claim 9, further comprising steps of: generating an error signal based on the output voltage and a reference voltage, generating a voltage command corresponding to the PWM signal based on the error signal, and generating the PWM signal based on the voltage command, generating a first current command corresponding to the effective value based on the voltage command and the input current, and generating a second current command corresponding to the input current based on the voltage command and the input current.

13. The method of operating the power converter as claimed in claim 12, further comprising steps of: determining whether the effective value is correspondingly lower than the current lower limit based on the first current command, and determining whether the input current is correspondingly greater than the predetermined threshold based on the second current command.

14. The method of operating the power converter as claimed in claim 12, further comprising steps of: detecting the output voltage to provide a first feedback voltage, and generating the error signal based on the first feedback voltage and the reference voltage, and detecting the input current to provide a second feedback voltage, and generating the first current command and the second current command based on the second feedback voltage and the voltage command.

15. The method of operating the power converter as claimed in claim 12, further comprising steps of: continuously generating the PWM signal based on the voltage command, and shielding the PWM signal by disabling the driver circuit.
